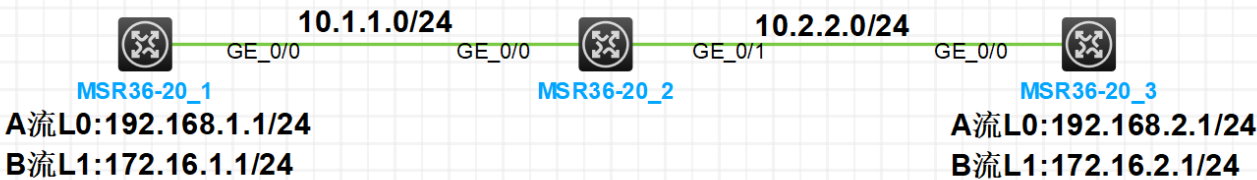


QOS实验

实验拓扑

实验需求

- 1.按照图示配置 IP 地址，R1 和 R3 分别配置环回口模拟业务网段
- 2.如图，网络中存在 A 流和 B 流两种业务流量
- 3.配置路由使全网互通
- 4.对 R1 发往 R3 的业务流提供 QOS 服务，R3 发往 R1 的业务流这里不做要求
- 5.在 R1 上对 A 流和 B 流进行分类，并标记 A 流 IP 优先级为 2，B 流优先级为 7
- 6.在 R2 上对 A 流和 B 流进行流量监管，承诺平均速率为 20M，超出平均速率的流量重标记 IP 优先级为 2
- 7.对流入 R3 优先级为 2 的流量提供快速转发服务，允许使用不超过 10M 的带宽，对优先级为 7 的流量提供确保转发服务，保证至少有 20M 的带宽可用



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7. 对流入 R3 优先级为 2 的流量提供快速转发服务，允许使用不超过 10M 的带宽，对优先级为 7 的流量提供确保转发服务，保证至少有 20M 的带宽可用

实验步骤

1. 配置 IP 地址部分略

```
1 [R1]int g0/0
2 [R1-GigabitEthernet0/0]ip ad 10.1.1.1 24
3 [R1-GigabitEthernet0/0]int l0
4 [R1-LoopBack0]ip ad 192.168.1.1 24
```

```

5 [R1-LoopBack0]int l1
6 [R1-LoopBack1]ip ad 172.16.1.1 24
7 ---
8 [R2]int g0/0
9 [R2-GigabitEthernet0/0]ip ad 10.1.1.2 24
10 [R2-GigabitEthernet0/0]int g0/1
11 [R2-GigabitEthernet0/1]ip ad 10.2.2.2 24
12 ---
13 [R3]int g 0/0
14 [R3-GigabitEthernet0/0]ip ad 10.2.2.3 24
15 [R3-GigabitEthernet0/0]int l0
16 [R3-LoopBack0]ip ad 192.168.2.1 24
17 [R3-LoopBack0]int l1
18 [R3-LoopBack1]ip ad 172.16.2.1 24

```

2. 配置路由部分略

```

1 [R1]rip
2 [R1-rip-1]v 2
3 [R1-rip-1]un s
4 [R1-rip-1]network 192.168.1.0
5 [R1-rip-1]network 172.16.1.0
6 [R1-rip-1]network 10.1.1.0
7 ---
8 [R2]rip
9 [R2-rip-1]v 2
10 [R2-rip-1]un s
11 [R2-rip-1]network 10.2.2.0
12 [R2-rip-1]network 10.1.1.1
13 [R2-rip-1]network 10.1.1.0
14 ---
15 [R3]rip
16 [R3-rip-1]v 2
17 [R3-rip-1]un s
18 [R3-rip-1]network 192.168.2.0
19 [R3-rip-1]network 172.16.2.0
20 [R3-rip-1]network 10.2.2.0

```

3. 在 R1 上对 A 流和 B 流进行分类，并标记 A 流 IP 优先级为 2，B 流优先级为 7

1. 在 R1 上创建 ACL 来匹配 A 流和 B 流流量

```

1 [R1]acl advanced 3000
2 [R1-acl-ipv4-adv-3000]rule permit ip source 192.168.1.0 0.0.0.255 destination 1
3 92.168.2.0 0.0.0.255
4
5 [R1]acl advanced 3001
6 [R1-acl-ipv4-adv-3001]rule permit ip source 172.16.1.0 0.0.0.255 destination 17
7 2.16.2.0 0.0.0.255
8
9

```

2. 在R1上创建类，关联对应ACL来对流量进行分类

```

1 [R1]traffic classifier a
2 [R1-classifier-a]if-match acl 3000
3
4 [R1]traffic classifier b
5 [R1-classifier-b]if-match acl 3001

```

3. 在R1上创建行为，对A流标记IP优先级为2，B流标记IP优先级为7

```

1 [R1]traffic behavior a
2 [R1-behavior-a]remark ip-precedence 2
3
4 [R1]traffic behavior b
5 [R1-behavior-b]remark ip-precedence 7
6

```

4. 在R1上创建QOS策略，绑定对应类和行为

```

1 [R1]qos policy yewu
2 [R1-qospolicy-yewu]classifier a behavior a
3 [R1-qospolicy-yewu]classifier b behavior b

```

5. 在R1的G0/0口的出方向下发QOS策略

```

1 [R1]int g0/0
2 [R1-GigabitEthernet0/0]qos apply policy yewu outbound

```

4. 在 R2 上对 A 流和 B 流进行流量监管，承诺平均速率为 20 M，超出平均速率的流量重标记 IP 优先级为 2

1. 在 R2 的 G0/0 接口的入方向上配置流量监管，Cir 20M，绿色放行，红色重标记 IP 优先级为 2

```
1 [R2-GigabitEthernet0/0]qos car inbound any cir 20000 green pass red remark-prec-pass 2
```

5. 对流入 R3 优先级为 2 的流量提供快速转发服务，允许使用不超过 10M 的带宽，对优先级为 7 的流量提供确保转发服务，保证至少有 20M 的带宽可用

1. 在 R2 上创建类，区分出 IP 优先级为 2 和 7 的流量

```
1 [R2]traffic classifier y2
2 [R2-classifier-y2]if-match ip-precedence 2
3 [R2]traffic classifier y7
4 [R2-classifier-y7]if-match ip-precedence 7
```

2. 在 R2 上创建行为，对 IP 优先级为 2 的流量提供 EF 服务，并设置最大可用带宽 10M，对 IP 优先级为 7 的流量提供 AF 服务，保证至少有 20M 的带宽可用

```
1 [R2]traffic behavior y2
2 [R2-behavior-y2]queue ef bandwidth 10000
3 [R2-behavior-y2]traffic behavior y7
4 [R2-behavior-y7]queue af bandwidth 20000
```

3. 在 R2 上创建 QOS 策略，绑定类和行为

```
1 [R2]qos policy yewu
2 [R2-qospolicy-yewu]classifier y2 behavior y2
3 [R2-qospolicy-yewu]classifier y7 behavior y7
```

4. 在 R2 的 G0/1 口的出方向下发 QOS 策略

```
1 [R2-qospolicy-yewu]int g0/1
2 [R2-GigabitEthernet0/1]qos apply policy yewu outbound
```

测试及查看

```
1 连通性测试
2 <R1>ping -a 192.168.1.1 192.168.2.1
```

```
3 Ping 192.168.2.1 (192.168.2.1) from 192.168.1.1: 56 data bytes, press CTRL+C to break
4 56 bytes from 192.168.2.1: icmp_seq=0 ttl=254 time=0.715 ms
5 56 bytes from 192.168.2.1: icmp_seq=1 ttl=254 time=0.780 ms
6
7 <R1>ping -a 172.16.1.1 172.16.2.1
8 Ping 172.16.2.1 (172.16.2.1) from 172.16.1.1: 56 data bytes, press CTRL+C to break
9 56 bytes from 172.16.2.1: icmp_seq=0 ttl=254 time=0.781 ms
10 56 bytes from 172.16.2.1: icmp_seq=1 ttl=254 time=0.719 ms
11
12 -----
13 查看接口qos策略信息
14 [R1]display qos policy interface g0/0
15 Interface: GigabitEthernet0/0
16   Direction: Outbound
17   Policy: yewu
18   Classifier: default-class
19     Matched : 51 (Packets) 4366 (Bytes)
20     5-minute statistics:
21       Forwarded: 0/22 (pps/bps)
22       Dropped : 0/0 (pps/bps)
23     Operator: AND
24     Rule(s) :
25       If-match any
26     Behavior: be
27     -none-
28   Classifier: a
29     Matched : 15 (Packets) 1470 (Bytes)
30     5-minute statistics:
31       Forwarded: 0/13 (pps/bps)
32       Dropped : 0/0 (pps/bps)
33     Operator: AND
34     Rule(s) :
35       If-match acl 3000
36     Behavior: a
37     Marking:
38       Remark ip-precedence 2
39   Classifier: b
40     Matched : 15 (Packets) 1470 (Bytes)
41     5-minute statistics:
42       Forwarded: 0/13 (pps/bps)
43       Dropped : 0/0 (pps/bps)
44     Operator: AND
45     Rule(s) :
46       If-match acl 3001
47     Behavior: b
48     Marking:
49       Remark ip-precedence 7
50
51 -----
52 [R2]display qos policy interface g0/1
53 Interface: GigabitEthernet0/1
54   Direction: Outbound
55   Policy: yewu
```

```
56 Classifier: default-class
57   Matched : 53 (Packets) 5578 (Bytes)
58   5-minute statistics:
59     Forwarded: 0/25 (pps/bps)
60     Dropped  : 0/0 (pps/bps)
61   Operator: AND
62   Rule(s) :
63     If-match any
64   Behavior: be
65   Default Queue:
66     Flow based Weighted Fair Queue:
67       Max number of hashed queues: 256
68       Matched  : 0 (Packets) 0 (Bytes)
69       Enqueued : 0 (Packets) 0 (Bytes)
70       Discarded: 0 (Packets) 0 (Bytes)
71       Discard Method: Tail
72 Classifier: y2
73   Matched : 10 (Packets) 980 (Bytes)
74   5-minute statistics:
75     Forwarded: 0/13 (pps/bps)
76     Dropped  : 0/0 (pps/bps)
77   Operator: AND
78   Rule(s) :
79     If-match ip-precedence 2
80   Behavior: y2
81   Expedited Forwarding:
82     Bandwidth 10000 (kbps) CBS 250000 (Bytes)
83     Matched  : 0 (Packets) 0 (Bytes)
84     Enqueued : 0 (Packets) 0 (Bytes)
85     Discarded: 0 (Packets) 0 (Bytes)
86 Classifier: y7
87   Matched : 15 (Packets) 1470 (Bytes)
88   5-minute statistics:
89     Forwarded: 0/13 (pps/bps)
90     Dropped  : 0/0 (pps/bps)
91   Operator: AND
92   Rule(s) :
93     If-match ip-precedence 7
94   Behavior: y7
95   Assured Forwarding:
96     Bandwidth 20000 (kbps)
97     Matched  : 0 (Packets) 0 (Bytes)
98     Enqueued : 0 (Packets) 0 (Bytes)
99     Discarded: 0 (Packets) 0 (Bytes)
100   Discard Method: Tail
```